UNIT 2 · TOPIC 2(D)

Fasteners, Keys and Shafts

Syllabus topic: Design of bearings, couplings, springs and fasteners

Bolts and threads

- **ISO metric property classes:** e.g., **8.8** → UTS = 8 × 100 = **800 MPa**; yield = 0.8 × 800 = **640 MPa**. Classes: 4.6, 5.8, 8.8, 10.9, 12.9. Match nuts by class number.
- **Thread designation:** M12 × 1.75 (nominal dia × pitch). Coarse threads are standard on farm equipment (they resist stripping in dirty/rusty service); fine threads are for adjustment.
- **Locking:** spring (split) washers, nyloc nuts, castle nut + split pin (wheel hubs), jam nuts, thread adhesives.

Keys and splines

Key	Features / use
Rectangular/square sunk key	Standard; w ≈ d/4
Woodruff	Self-aligning, semicircular — tapered shafts, magnetos
Feather key	Axial sliding
Taper key + gib head	—
Splines	Heavy sliding torque — PTO shaft: e.g., 6-spline 540 rpm, 21-spline 1000 rpm

- Design keys in **shear and crushing**. A keyway weakens the shaft — a **~25%** strength reduction is a typical design allowance.

Shafts

$$\tau = 16T/(\pi d^3)\ \text{(torsion)}$$

$$T_e = \sqrt{(M^2 + T^2)}\ \text{(combined bending + torsion, max-shear-stress theory)}$$

- Use standard transmission-shaft sizes.
- Check the critical (whirling) speed of long shafts.
- Rivets and welds replace bolts in frames; weld strength design is in Unit 8.

Practice MCQs

Q38. A bolt of property class 8.8 has minimum ultimate tensile strength and yield strength of:

(a) 800 and 640 MPa
(b) 880 and 800 MPa
(c) 800 and 880 MPa
(d) 640 and 800 MPa

Q39. The 540-rpm tractor PTO shaft has:

(a) 21 splines
(b) 6 splines
(c) 8 splines
(d) 10 splines

Q40. The self-aligning key seated in a semicircular shaft recess is the:

(a) Feather key
(b) Kennedy key
(c) Woodruff key
(d) Gib-head key

Answers

Q38 — (a) 800 and 640 MPa — *Class X.Y → UTS = 100X MPa; yield = 0.Y × UTS.*

Q39 — (b) 6 splines

Q40 — (c) Woodruff key

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 4 (Elements of Rotary Power-Transmission Systems) and Appendix (ASAE V-belt standards); standard machine design references (Shigley; V.B. Bhandari); ASAE/IS standards.